

* Majority elements -;

Example -;

$$a[6] = 1 \ 2 \ 6 \ 1 \ 1 \ 1$$

$$\text{freq}(1) = 4 > \frac{6}{2} \quad \text{ans} = 1$$

Moore's voting algorithm

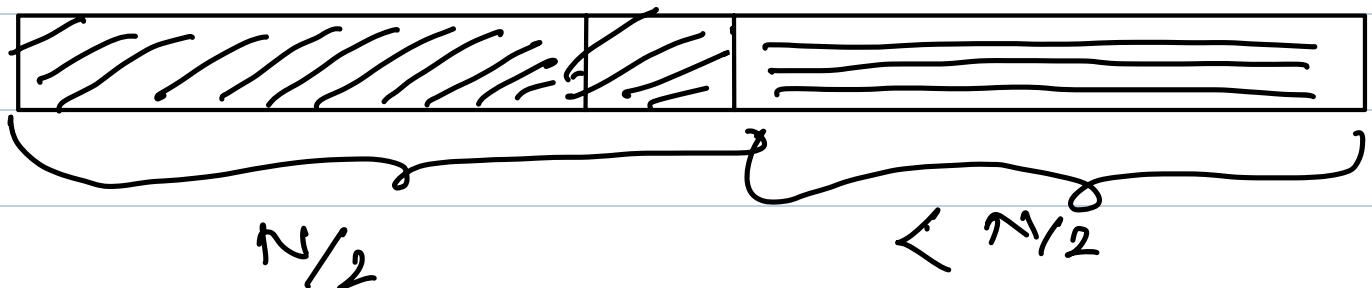
$$a[9] = 3 \ 4 \ 4 \ 8 \ 4 \ 9 \ 4 \ 3 \ 4$$

$$\text{freq}(4) = 5 > \frac{9}{2} = \text{ans} = 4$$

Idea -;

Majority element $\rightarrow \text{freq} > \frac{N}{2}$

only 1 majority element can exist



* Observation:-

- 1) At max, we can have 1 majority element.
 - 2) freq of majority element will be greater than all other elements combined.
- majority remains same.

If majority element is not assured
run loop to check if elements has
 $\text{freq} > N/2$

Moore's Voting Algorithm

TC: $O(N)$

SC: $O(1)$

* Sum of Even Indexed elements:-

make prefix array with even
Indexed sum.

TC: $O(2N)$, SC: $O(N)$

* Special Index:-

$$\text{Sum of all Even Index} = \text{Sum of all odd index}$$

Generalisation:-

$$A = a_0, a_1, a_2, \dots, a_{i-1}, a_i, a_{i+1}, \dots, a_{n-1}$$

$$S_{\text{even}} = \text{SumEven}[0, i-1] + \text{Sumodd}[i+1, N-1]$$

$$S_{\text{odd}} = \text{Sumodd}[0, i-1] + \text{SumEven}[i+1, N-1]$$

1) Construct preEven & preodd

$$\text{SumEven}[0, i-1] = \text{preEven}[i-1]$$

$$\text{Sumodd}[0, i-1] = \text{preodd}[i-1]$$

$$\text{Sumodd}[i+1, N-1] = \text{preodd}[N-1] - \text{preodd}[i]$$

$$\text{SumEven}[i+1, N-1] =$$

$$\text{preEven}[N-1] - \text{preEven}[i]$$

$$TC: O(N)$$

$$SC: O(N)$$